

TECHSHIP INTERNSHIP

Day 3 – Functions, Arrays, Objects & ES6 concepts

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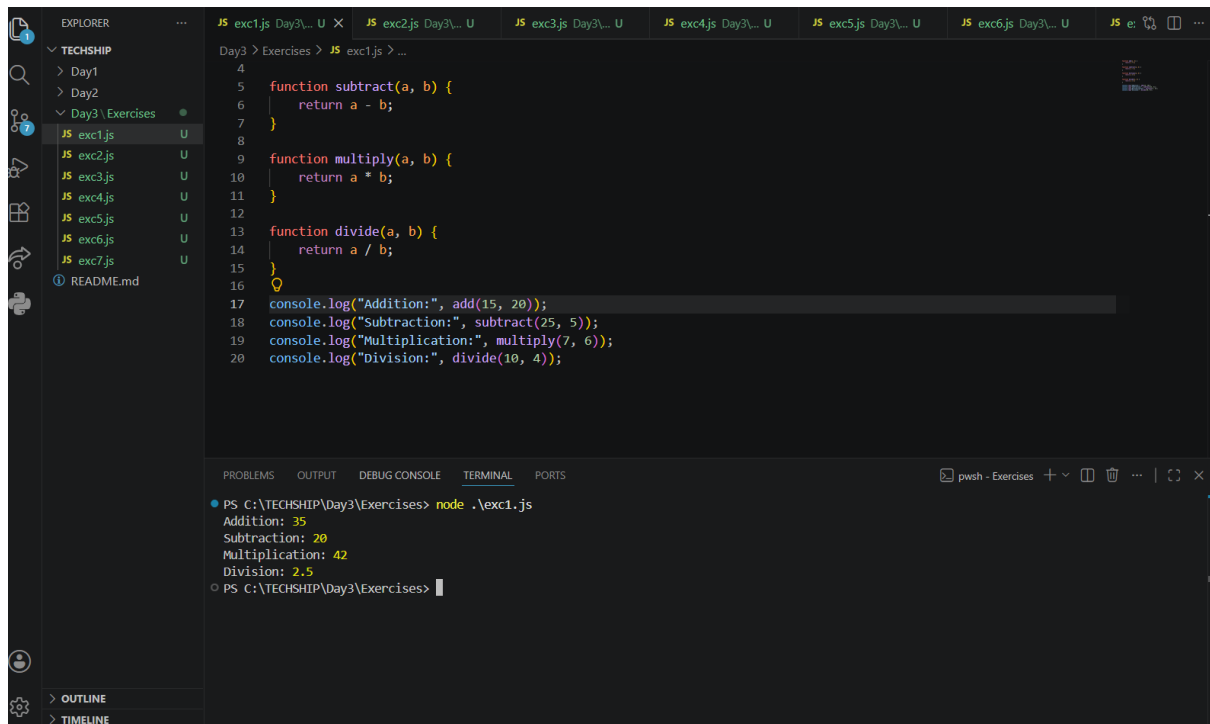
On Day 3 of the TECHSHIP Internship Program, I learned about JavaScript functions, arrays, objects, and ES6 features. I understood how functions help in writing reusable code and learned the difference between normal functions and arrow functions. I also worked with arrays and practiced using array methods such as `forEach()`, `map()`, and `filter()` to process data efficiently.

I learned how objects are used to store related information and how their properties can be accessed and updated. In addition, I explored important ES6 features including template literals, destructuring, and the spread operator. These features make JavaScript code more readable and easier to write.

The ES6 feature I found most useful was the spread operator because it allows copying and modifying arrays or objects without changing the original data. During the implementation, I faced some difficulty understanding destructuring syntax and applying array methods correctly, but after practicing the exercises and assignment, I was able to understand these concepts better.

Exercises

Exercise 1 – Function Based Calculator



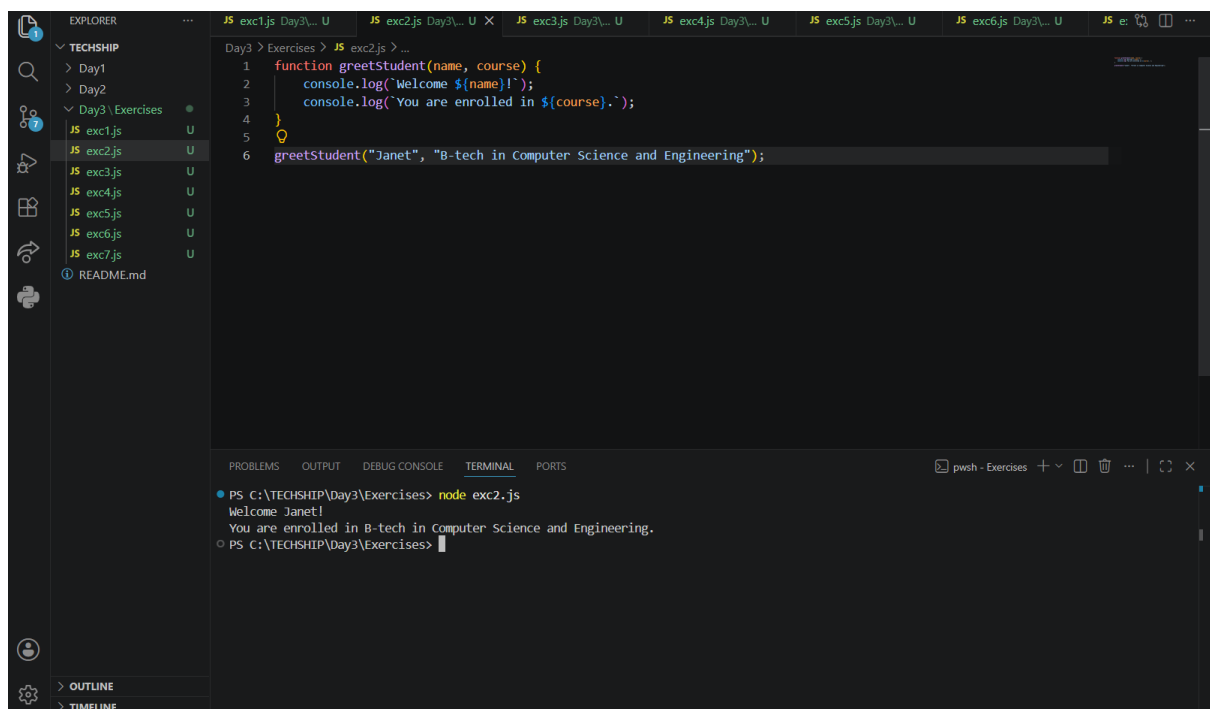
The screenshot shows a VS Code editor with a project named 'TECHSHIP'. The Explorer sidebar shows a folder 'Day3' containing a subfolder 'Exercises' with files 'exc1.js' through 'exc7.js'. The main editor displays 'exc1.js' with the following code:

```
4
5 function subtract(a, b) {
6   return a - b;
7 }
8
9 function multiply(a, b) {
10  return a * b;
11 }
12
13 function divide(a, b) {
14  return a / b;
15 }
16
17 console.log("Addition:", add(15, 20));
18 console.log("Subtraction:", subtract(25, 5));
19 console.log("Multiplication:", multiply(7, 6));
20 console.log("Division:", divide(10, 4));
```

The bottom panel shows the 'TERMINAL' tab with the following output:

```
PS C:\TECHSHIP\Day3\Exercises> node .\exc1.js
Addition: 35
Subtraction: 20
Multiplication: 42
Division: 2.5
PS C:\TECHSHIP\Day3\Exercises>
```

Exercise 2 – Student Greeting Function



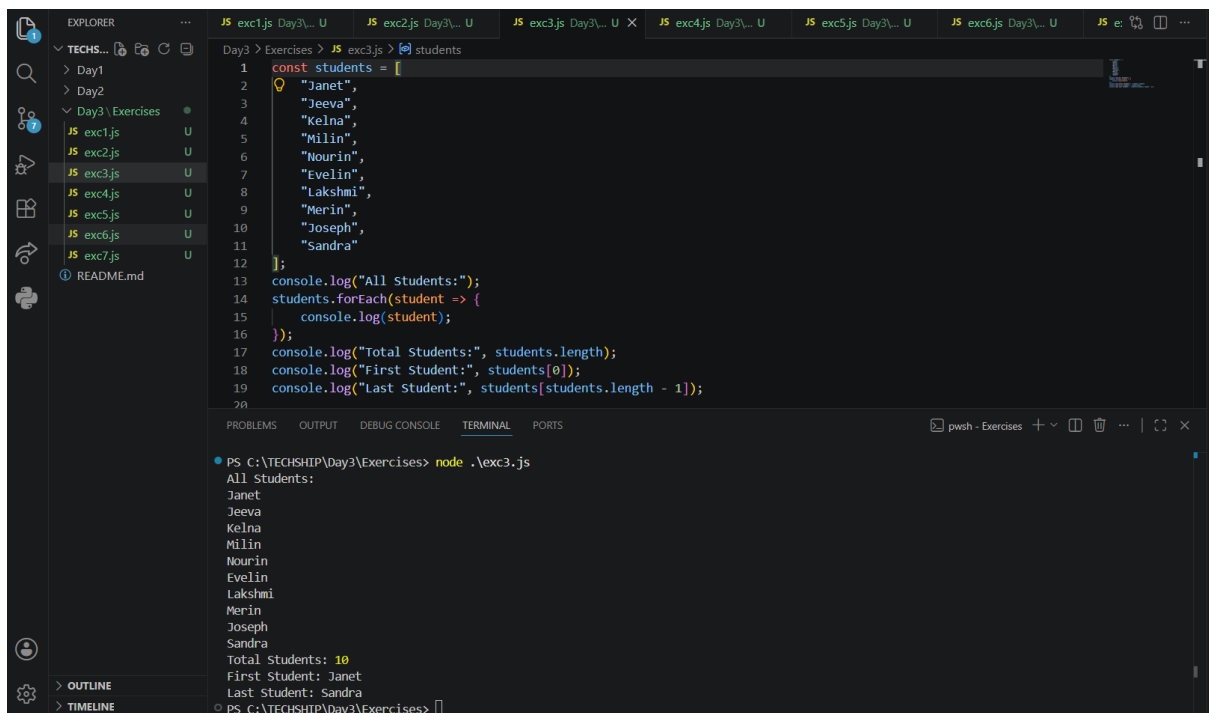
The screenshot shows the same VS Code editor with the 'Exercises' folder. The main editor displays 'exc2.js' with the following code:

```
1 function greetStudent(name, course) {
2   console.log(`Welcome ${name}!`);
3   console.log(`You are enrolled in ${course}.`);
4 }
5
6 greetStudent("Janet", "B-tech in Computer Science and Engineering");
```

The bottom panel shows the 'TERMINAL' tab with the following output:

```
PS C:\TECHSHIP\Day3\Exercises> node exc2.js
Welcome Janet!
You are enrolled in B-tech in Computer Science and Engineering.
PS C:\TECHSHIP\Day3\Exercises>
```

Exercise 3 – Working with Arrays



```
Day3 > Exercises > JS exc3.js > students
1  const students = [
2      "Janet",
3      "Jeeva",
4      "Kelna",
5      "Milin",
6      "Nourin",
7      "Evelin",
8      "Lakshmi",
9      "Merin",
10     "Joseph",
11     "Sandra"
12 ];
13 console.log("All Students:");
14 students.forEach(student => {
15     console.log(student);
16 });
17 console.log("Total Students:", students.length);
18 console.log("First Student:", students[0]);
19 console.log("Last Student:", students[students.length - 1]);
20
```

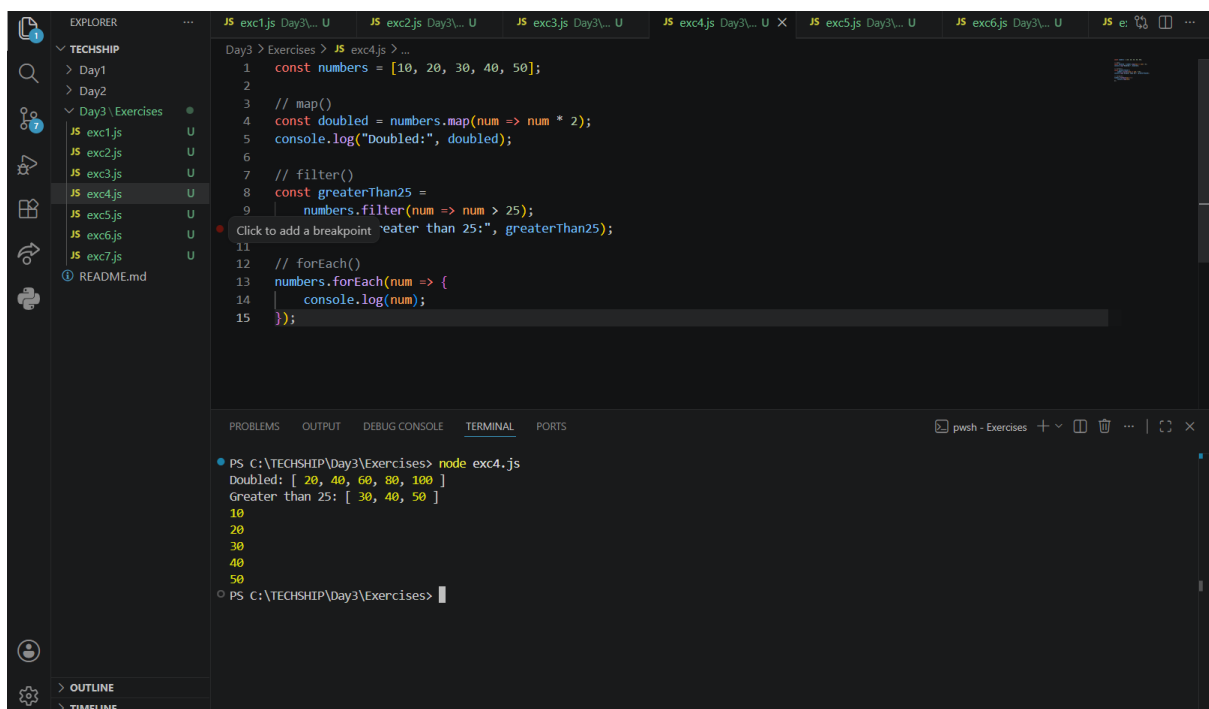
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\TECHSHIP\Day3\Exercises> node .\exc3.js

All Students:
Janet
Jeeva
Kelna
Milin
Nourin
Evelin
Lakshmi
Merin
Joseph
Sandra
Total Students: 10
First Student: Janet
Last Student: Sandra

PS C:\TECHSHIP\Day3\Exercises>

Exercise 4 – Array Methods



```
Day3 > Exercises > JS exc4.js > ...
1  const numbers = [10, 20, 30, 40, 50];
2
3  // map()
4  const doubled = numbers.map(num => num * 2);
5  console.log("Doubled:", doubled);
6
7  // filter()
8  const greaterThan25 =
9      numbers.filter(num => num > 25);
10
11  // Click to add a breakpoint "reater than 25:", greaterThan25);
12
13  // forEach()
14  numbers.forEach(num => {
15      console.log(num);
16  });
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\TECHSHIP\Day3\Exercises> node exc4.js

Doubled: [20, 40, 60, 80, 100]
Greater than 25: [30, 40, 50]
10
20
30
40
50

PS C:\TECHSHIP\Day3\Exercises>

Exercise 5 – Student Objects

The screenshot shows the Visual Studio Code editor with a project named 'TECHSHIP'. The Explorer sidebar on the left shows a folder structure with 'Day3' containing an 'Exercises' folder. Inside 'Exercises', there are several JavaScript files: exc1.js, exc2.js, exc3.js, exc4.js, exc5.js (selected), exc6.js, and exc7.js, along with a README.md file. The main editor window displays the code for exc5.js:

```
1  const student = {
2    name: "Janet",
3    age: 21,
4    college: "Aisat Engineering college",
5    course: "BTech",
6    skills: ["HTML", "CSS", "JavaScript"]
7  };
8
9  console.log(student.name);
10 console.log(student.age);
11 console.log(student.college);
12 console.log(student.course);
13 console.log(student.skills);
```

Below the editor, the TERMINAL panel shows the output of running 'node exc5.js' in a PowerShell window:

```
PS C:\TECHSHIP\Day3\Exercises> node exc5.js
Janet
21
Aisat Engineering college
BTech
[ 'HTML', 'CSS', 'JavaScript' ]
PS C:\TECHSHIP\Day3\Exercises>
```

Exercise 6 – Employee Objects

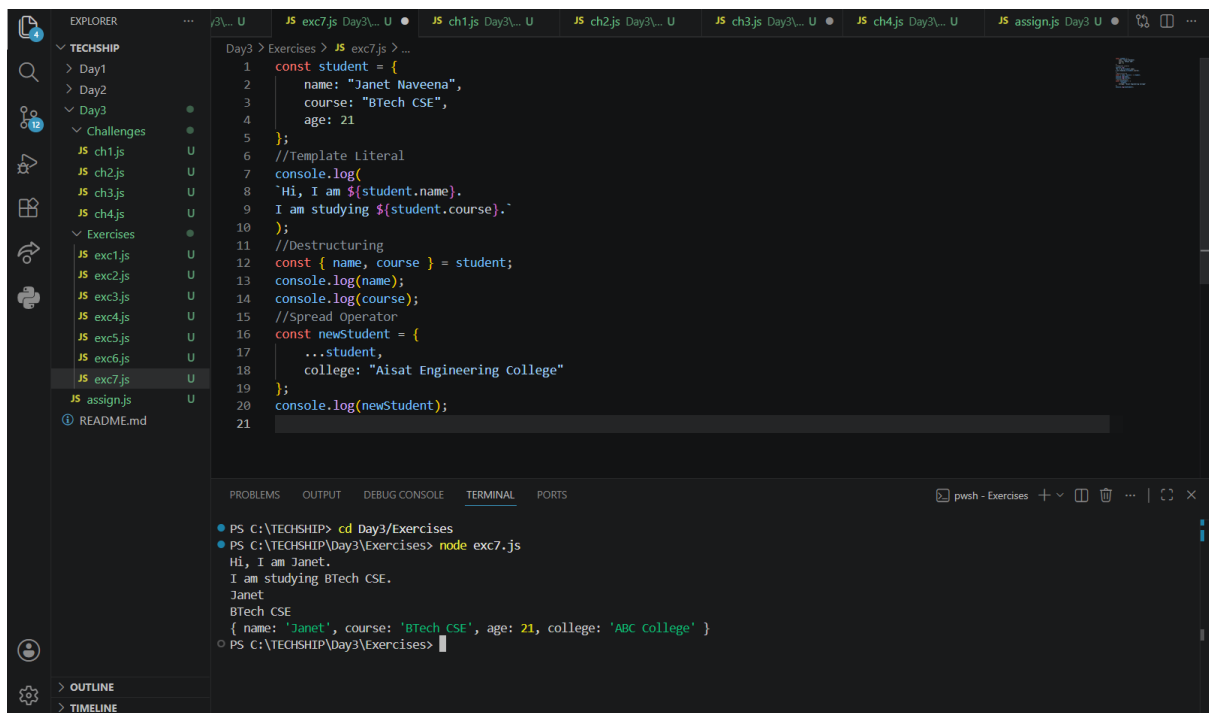
The screenshot shows the Visual Studio Code editor with the same 'TECHSHIP' project. The Explorer sidebar is identical to the previous screenshot. The main editor window displays the code for exc6.js:

```
1  const employee = {
2    employeeId: 1324,
3    name: "Janet",
4    department: "Marketing",
5    salary: 50000
6  };
7
8  console.log("Before Update:");
9  console.log(employee);
10
11 // Update salary
12 employee.salary = 90000;
13
14 console.log("After Update:");
15 console.log(employee);
16
```

The TERMINAL panel shows the output of running 'node exc6.js' in a PowerShell window:

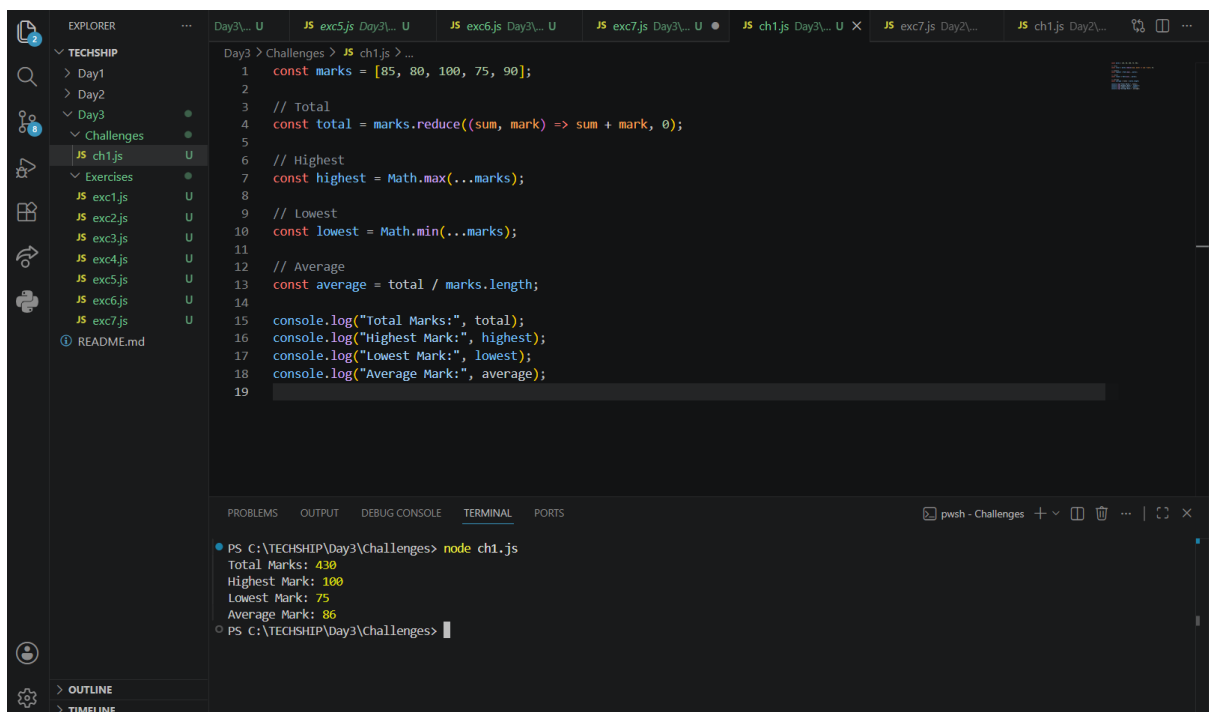
```
PS C:\TECHSHIP\Day3\Exercises> node exc6.js
Before Update:
{
  employeeId: 1324,
  name: 'Janet',
  department: 'Marketing',
  salary: 50000
}
After Update:
{
  employeeId: 1324,
  name: 'Janet',
  department: 'Marketing',
  salary: 90000
}
PS C:\TECHSHIP\Day3\Exercises>
```

Exercise 7 – ES6 Practice

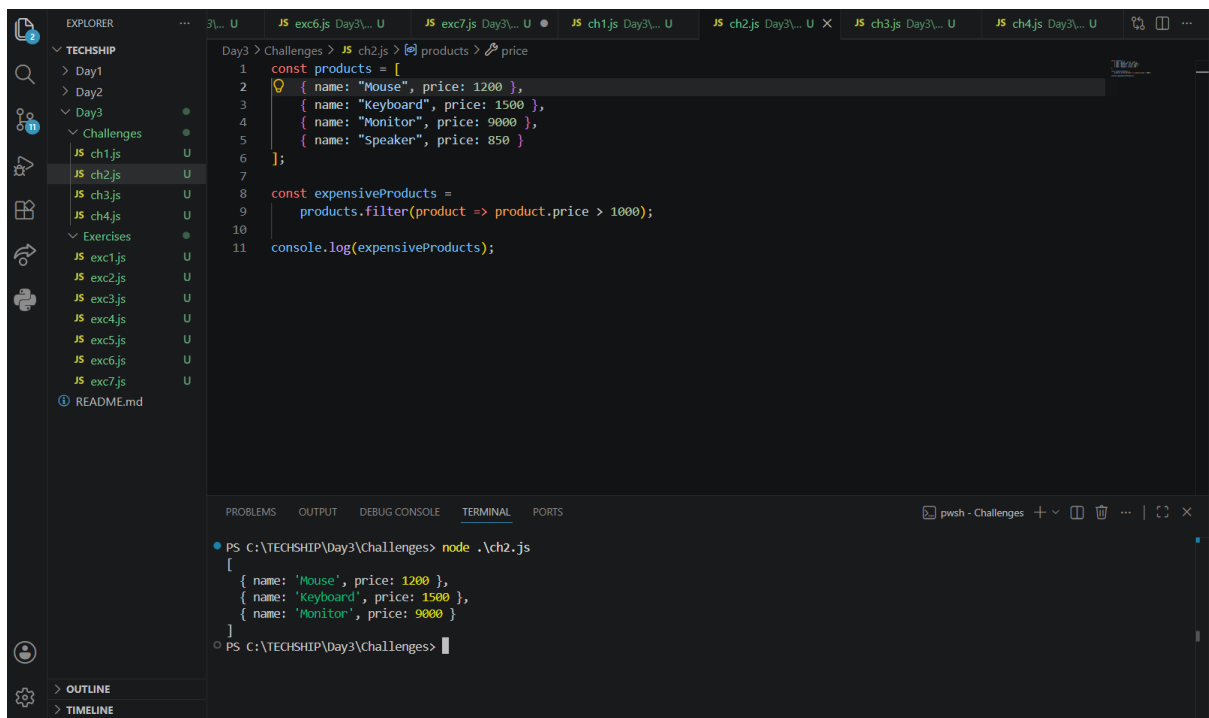


Challenge Problems

Challenge 1 – Student Marks Analyzer



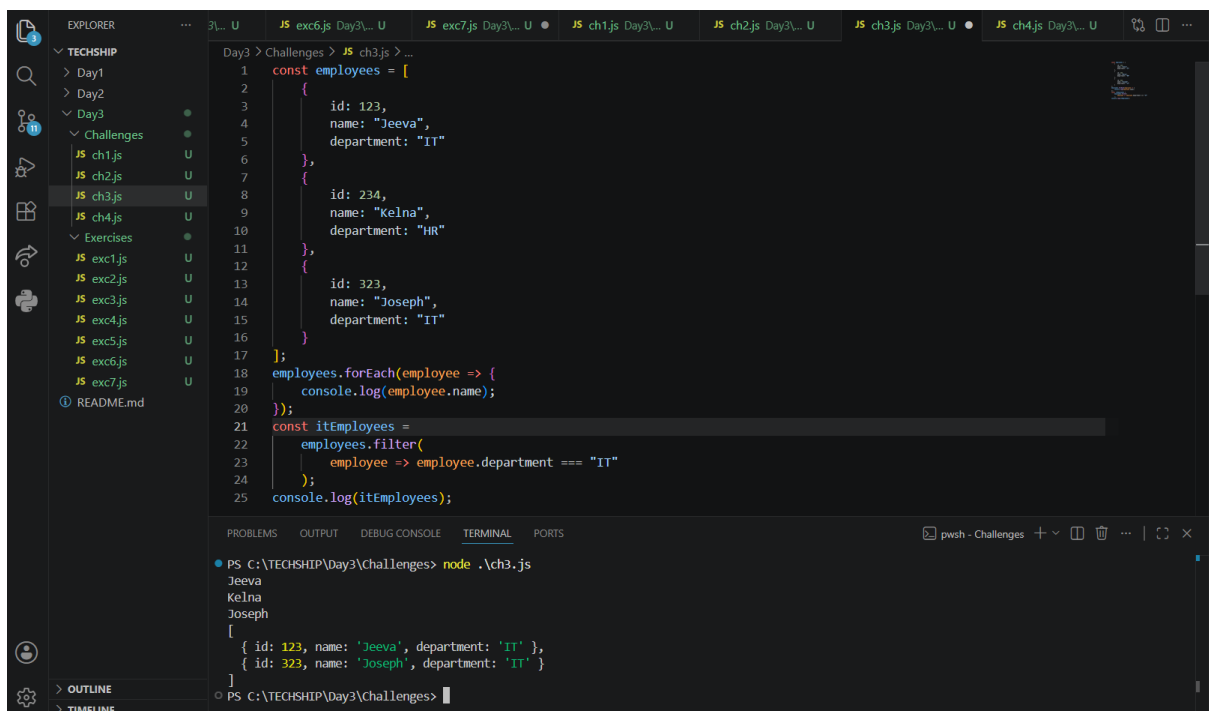
Challenge 2 – Product Filter System



```
Day3 > Challenges > JS ch2.js > products > price
1  const products = [
2    { name: "Mouse", price: 1200 },
3    { name: "Keyboard", price: 1500 },
4    { name: "Monitor", price: 9000 },
5    { name: "Speaker", price: 850 }
6  ];
7
8  const expensiveProducts =
9    products.filter(product => product.price > 1000);
10
11 console.log(expensiveProducts);
```

```
PS C:\TECHSHIP\Day3\Challenges> node .\ch2.js
[
  { name: 'Keyboard', price: 1500 },
  { name: 'Monitor', price: 9000 }
]
```

Challenge 3 – Employee Directory



```
Day3 > Challenges > JS ch3.js > ...
1  const employees = [
2    {
3      id: 123,
4      name: "Jeeva",
5      department: "IT"
6    },
7    {
8      id: 234,
9      name: "Kelna",
10     department: "HR"
11   },
12   {
13     id: 323,
14     name: "Joseph",
15     department: "IT"
16   }
17 ];
18 employees.forEach(employee => {
19   console.log(employee.name);
20 });
21 const itEmployees =
22   employees.filter(
23     employee => employee.department === "IT"
24   );
25 console.log(itEmployees);
```

```
PS C:\TECHSHIP\Day3\challenges> node .\ch3.js
Jeeva
Kelna
Joseph
[
  { id: 123, name: 'Jeeva', department: 'IT' },
  { id: 323, name: 'Joseph', department: 'IT' }
]
```

Challenge 4 – Course Enrollment System

The screenshot shows the VS Code interface with the following details:

- Explorer View:**
 - TECHSHIP
 - Day1
 - Day2
 - Day3
 - Challenges
 - ch1.js
 - ch2.js
 - ch3.js
 - ch4.js (selected)
 - Exercises
 - exc1.js
 - exc2.js
 - exc3.js
 - exc4.js
 - exc5.js
 - exc6.js
 - exc7.js
 - README.md

- Source Control View:**
- ch4.js (staged)
- Run and Debug View:**
- Command: `node .\ch4.js`
- Output:


```
Student Profile
-----
Name: Janet
Age: 21
Course: BTech CSE
Skills: HTML, CSS, JavaScript
```

Assignment

Student Management System

```
Day3 > JS assign.js > students
1  const students = [
2    {id: 111, name: "Evelin", age: 20, course: "BCA", marks: [75, 80, 60, 85, 90]},
3    {id: 112, name: "Anna", age: 21, course: "BSc CS", marks: [50, 66, 65, 70, 80]},
4    {id: 113, name: "Janet", age: 21, course: "BTech CSE", marks: [90, 95, 92, 88, 91]},
5    {id: 114, name: "Kelna", age: 22, course: "BTech CSE", marks: [75, 70, 68, 72, 80]},
6    {id: 115, name: "Diya", age: 21, course: "BCom", marks: [78, 80, 75, 82, 79]}
7  ];
8  // Calculate average for each student
9  const studentAverages = students.map(student => {
10    const total = student.marks.reduce(
11      (sum, mark) => sum + mark,
12      0
13    );
14    const average = total / student.marks.length;
15    return {
16      ...student,
17      average: average
18    };
19  });
20  // Students scoring above 75%
21  const topStudents = studentAverages.filter(
22    student => student.average > 75
23  );
24  // Find top performer
25  const topper = studentAverages.reduce(
26    (highest, student) =>
27      student.average > highest.average
28      ? student
29      : highest
30  );
31  console.log("=====");
32  console.log("STUDENT MANAGEMENT REPORT");
33  console.log("=====");
34  // Display all students with report
35  studentAverages.forEach(student => {
36    let status;
37    if (student.average >= 80) {
38      status = "Excellent";
39    } else if (student.average >= 70) {
40      status = "Good";
41    } else {
42      status = "Needs Improvement";
43    }
44    console.log(`ID: ${student.id}
45    Name: ${student.name}
46    Course: ${student.course}
47    Average Marks: ${student.average.toFixed(2)}
48    Status: ${status}`);
49  });
50  // Students above 75%
51  console.log("-----");
52  console.log("Students Scoring Above 75%:");
53
54  topStudents.forEach(student => {
55    console.log(
56      `${student.name} - ${student.average.toFixed(2)}
57    );
58  });
59  // Top performer
60  console.log("-----");
61  console.log(`Top Performer: ${topper.name}`);
62  console.log(`Highest Mark: ${topper.average.toFixed(2)}`);
63  console.log("=====");
```


The screenshot shows the Visual Studio Code interface with the following components:

- EXPLORER:** A file tree on the left showing a project named 'TECHSHIP'. It contains folders 'Day1', 'Day2', and 'Day3'. 'Day3' contains subfolders 'Challenges' and 'Exercises'. 'Challenges' has files 'ch1.js', 'ch2.js', 'ch3.js', and 'ch4.js'. 'Exercises' has files 'exc1.js', 'exc2.js', 'exc3.js', 'exc4.js', 'exc5.js', 'exc6.js', and 'exc7.js'. The file 'assign.js' is selected in the 'Exercises' folder. A 'README.md' file is also visible.
- TERMINAL:** The main panel displays the output of running 'node assign.js'. The output is a 'STUDENT MANAGEMENT REPORT' listing student details for IDs 111 through 115. It includes names, courses, average marks, and status. A summary section lists students scoring above 75% (Evelin, Janet, Diya) and identifies Janet as the top performer with a highest mark of 91.20.
- PROBLEMS:** A panel on the left of the terminal showing the command 'PS C:\TECHSHIP\Day3> node assign.js'.
- OUTPUT:** A panel on the right of the terminal showing the command 'pwsh Day3'.

```
PS C:\TECHSHIP\Day3> node assign.js
=====
STUDENT MANAGEMENT REPORT
=====
ID: 111
Name: Evelin
Course: BCA
Average Marks: 78.00
Status: Good
ID: 112
Name: Anna
Course: BSc CS
Average Marks: 66.20
Status: Needs Improvement
ID: 113
Name: Janet
Course: BTech CSE
Average Marks: 91.20
Status: Excellent
ID: 114
Name: Kelna
Course: BTech CSE
Average Marks: 73.00
Status: Good
ID: 115
Name: Diya
Course: BCom
Average Marks: 78.80
Status: Good
-----
Students Scoring Above 75%:
Evelin - 78.00
Janet - 91.20
Diya - 78.80
-----
Top Performer: Janet
Highest Mark: 91.20
=====
```